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Comparison of FilterFFPE's one-step and two-step procedures for improving SV calling in FFPE samples

Introduction

Next-generation sequencing data from formalin-fixed paraffin-embedded (FFPE) samples is enriched with artifact chimeric reads[1] . These reads are generated during the sequencing process[1], rather than formed due to real structural variants (SV). However, existing SV detection tools cannot distinguish between these two kinds of chimeric reads, thus resulting in a large number of false positive SV calls. To specifically remove artifact chimeric reads, we developed FilterFFPE[2], a two-step filtering algorithm. Briefly, the first filtration step is based on the number of reads that share the same breakpoint, and the second on the presence of short complementary sequences around the breakpoint. While the first step identifies all possible artifact chimeric reads, the optional second step is specifically designed for samples with low coverage and/or low SV frequency. To access and compare the effect of these two steps, we evaluated the performance of FilterFFPE's one-step and two-step procedures on improving SV calling with 41 simulated FFPE samples.

Methods

To access the performance of FilterFFPE, we simulated 41 samples mimicking FFPE whole-genome sequencing data with SimFFPE[3]. Four hundred SVs were simulated using RSVsim[4]. Altogether, we generated three simulated data sets with varying coverage, SV frequencies and levels of artifacts (Table 1). FilterFFPE was applied to the simulated data with one-step and two-step filtration. Delly[5] and Lumpy[6] were used to call SVs. The performance on SV calling for data with no filter, one-step and two-step filtration was evaluated and compared.

Table 1: Characteristics of simulated data sets. Abbreviations: SV - structural variant; ACF - artifact chimeric fragment.

Name	n	Coverage	SV frequency	Proportion of ACFs
Sim1	10	10-100x	50%	10%
Sim2	10	50x	10-100%	10%
Sim3	21	50x	50%	0-20%

Results

Our results show that FilterFFPE substantially improves the precision of SV calling for simulated samples with almost constant sensitivity (Figure 1a). For samples with low coverage or low SV frequency, sensitivity decreased slightly after application of FilterFFPE's one-step filtration, and recovered to some extent after application of the additional second step (Figure 1b).

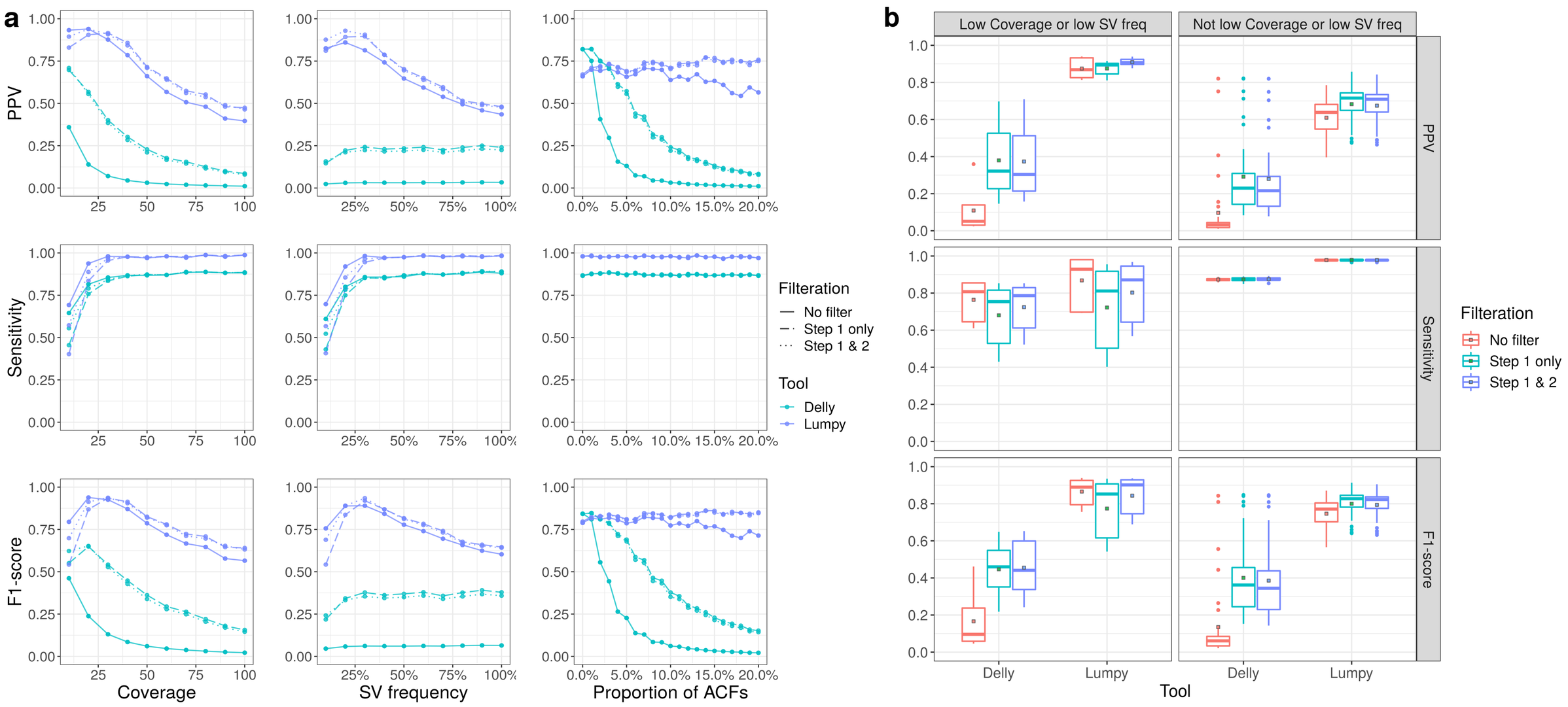

Figure 1: (a) Performance on SV calling for our three simulated data sets with no filter, one-step and two-step filtration. (b) Performance on SV calling for simulated samples with low coverage or low SV frequency (n=6) and other samples (n=35). Results for data with no filter, one-step and two-step filtration are shown. Abbreviations: PPV - positive predictive value; freq - frequency.

Conclusion

Compared to one-step filtration, the second filtration step of FilterFFPE ensures that non-artifact chimeric reads are kept, thus leading to higher sensitivity for SV calling. We recommend the two-step procedure of FilterFFPE in cases of low sample coverage or heterogeneous samples.

References

[1] Haile S. et al. Sources of erroneous sequences and artifact chimeric reads in next generation sequencing of genomic DNA from formalin-fixed paraffin-embedded samples. *Nucleic Acids Res.* 2019, 47(2):e12. **[2]** DOI: 10.18129/B9.bioc.FilterFFPE **[3]** DOI: 10.18129/B9.bioc.SimFFPE **[4]** Bartenhagen C, Dugas M. RSVSim: an R/Bioconductor package for the simulation of structural variations. *Bioinformatics* 2013;29(13):1679-1681. **[5]** Rausch T. et al. DELLY: structural variant discovery by integrated paired-end and split-read analysis. *Bioinformatics* 2012, 28(18):i333-i339. **[6]** Layer RM. et al. LUMPY: a probabilistic framework for structural variant discovery. *Genome Biol.* 2014, 15(6):R84.

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